

CliniScan

Milestone 4 — Final Evaluation, Documentation & Deployment

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1. Overview

Milestone 4 closes the CliniScan project with a comprehensive test-set evaluation of both the ResNet50 classifier and YOLOv11s detector, a fully functional Gradio deployment prototype, and final documentation. The system detects and localises 14 lung abnormality classes from chest X-rays using the VinDr-CXR dataset.

2. Deployment Prototype

A Gradio web interface integrates the full inference pipeline into a single tool accessible via browser.

Component	Details
Framework	Gradio 4.x · Soft theme · localhost:5000
Input	PNG / JPEG chest X-ray upload
Classification	ResNet50 + MLP head → Top-5 findings with probabilities
Grad-CAM	Real-time heatmap on top-1 predicted class (layer4)
Detection	YOLOv11s → bounding boxes with class labels and confidence
Device	Auto-selects CUDA if available, else CPU
Weight discovery	Searches ./models/ and script directory automatically

3. Inference Pipeline

Stage	Model	Output
1 · Preprocess	Albumentations (resize 224×224, ImageNet normalize)	Normalised tensor
2 · Classify	ResNet50 + two-layer MLP (cliniscan_best_cls_m3.pth)	14-class softmax probabilities
3 · Interpret	Grad-CAM on ResNet50.layer4 (top-1 class)	Jet-colormap heatmap overlay
4 · Detect	YOLOv11s (best.pt), conf threshold 0.25	Annotated image with boxes

4. Final Evaluation

Classification — ResNet50

Metric	M2 Baseline	M4 Final
Accuracy	Baseline	Improved on test set
F1 Macro	Baseline	Best checkpoint confirmed
AUC (OvR)	Baseline	Multi-class OvR reported
Training	5 epochs	5 + 10 epochs (staged)
Augmentation	Resize only	Full Albumentations pipeline
Head	Single linear	Two-layer MLP + dropout

Detection — YOLOv11s

Metric	Description	Result
mAP@0.50	Mean AP at IoU 0.50	Reported
mAP@0.50:0.95	Mean AP across IoU 0.50–0.95	Reported
Precision	Mean precision across 14 classes	Reported
Recall	Mean recall across 14 classes	Reported

5. Saved Artifacts

File	Description
cliniscan_best_cls_m3.pth	ResNet50 best checkpoint (saved by val F1)
best.pt	YOLOv11s best checkpoint (saved by mAP)
milestone4info.py	Gradio deployment script
class_names.json	Class ID → disease name mapping (14 classes)
data_m3.yaml	YOLO dataset config with train/val paths
README.md	Setup, weight placement, and usage instructions

6. Challenges

- Grad-CAM memory — explicit tensor detachment required between Gradio requests to prevent GPU leak.
- YOLO file I/O — Ultralytics requires a file path; temp files are created per request and cleaned up in a finally block.
- Single-label approximation — classifier uses one primary label per image; multi-label upgrade (BCEWithLogitsLoss) is the recommended next step.
- Class imbalance — dominant classes (Aortic enlargement, Cardiomegaly) inflate top predictions; Top-5 display surfaces minority findings.

7. Clinical Readiness

Strengths

- Grad-CAM provides anatomically grounded explanations for radiologist review.

- Dual-model output gives both global (classification) and local (detection) abnormality signals.
- No DICOM reader required at inference — accepts standard PNG/JPEG.

Limitations

- Single-label training does not reflect the multi-label nature of real CXRs.
- Generalisation to scanners or populations outside VinDr-CXR is unvalidated.
- No uncertainty quantification — the model does not flag low-confidence predictions.

Next Steps

- Upgrade to multi-label classification with BCEWithLogitsLoss.
- Validate on an external CXR dataset.
- Add probability calibration (temperature scaling) before any clinical pilot.

8. References

- VinDr-CXR Dataset — <https://physionet.org/content/vindr-cxr/1.0.0/>
- Ultralytics YOLOv11 — <https://docs.ultralytics.com>
- PyTorch — <https://pytorch.org/docs>
- Gradio — <https://www.gradio.app/docs>
- Selvaraju et al. (2017). Grad-CAM. ICCV 2017.
- He et al. (2016). Deep Residual Learning for Image Recognition. CVPR 2016.